NATTAPAT (VERN) PRAYOONTHONG

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Education

Carnegie Mellon University, Carnegie Institute of Technology — Pittsburgh, PA
B.S. in Mechanical Engineering, Minor in Robotics (3.45 GPA) — *Expected May 2028*

Relevant Experience

CMU Lunabotics (CAD Design, DFM, GD&T, Manual Machining, 3D Printing, CNC) — Pittsburgh, PA
Mobility Subteam Lead — *August 2025 - Present*

- **Lead** a 6-member mobility subteam designing, manufacturing the tracked drive system for an 80 kg lunar rover
- **Coordinated** the transition of 20+ CAD components from Onshape to SolidWorks across a 6-member mobility team.
- **Designed and integrated** a 6-tooth, 5.4-in-diameter drive sprocket and 9 mechanical interfaces through 4 iterative redesigns, incorporating increasing DFM considerations to improve manufacturability, assembly, and fit
- **Manufactured** 60+ aluminum tread plates using precision cutting and CNC drilling
- **Fabricated and integrated 10+ track components** using 3D printing, manual milling, and hands-on mechanical assembly for the final rover

National Science and Technology Development Agency (OpenCV, Mediapipe, Python, YOLO) — Bangkok, Thailand
Intern — *June 2025 - July 2025*

- **Developed and tested** an automated object-counting system using image thresholding and contour detection
- **Built and annotated** a custom dataset and trained a YOLOv11-OBB model for oriented object detection
- **Analyzed** performance across lighting and orientation conditions to identify failure cases and improve system robustness

Engineering Projects

Hammock Chair Design (FEA, Prototyping, Mechanical Design) — Pittsburgh, PA
March 2026 - May 2026

- **Designed, fabricated, and tested** a portable load-bearing chair and rope-support system with a 4-person team
- **Performed** hand calculations to determine structural loads and guide component sizing and mechanical design
- **Conducted FEA** to evaluate stress and deformation of load-bearing components, achieving a simulated factor of safety of 2.52
- **Built and tested** the final prototype, successfully validating stability and structural performance under loads up to 200 lb

General Robotics (Robotics, CAD, SolidWorks, 3D printing, PID) — Pittsburgh, PA
January 2026 - May 2026

- **Designed and fabricated** mechanical components for an Urban Search and Rescue robot using SolidWorks
- **Designed** a custom mold for casting polyurethane wheels, incorporating geometry, fit, and manufacturability constraints
- **Built, tested, and iterated** robot design through rapid prototyping and performance evaluation
- **Developed and tuned** PID control algorithms for a self-balancing robot through repeated testing and data analysis

Posture Your Focus (Python, OpenCV, MediaPipe) — Pittsburgh, PA
April 2025 - May 2025

- **Developed** a real-time computer vision application using Python, OpenCV, and MediaPipe to monitor 4 user behaviors, including slouching, screen distance, phone use, and inactivity
- **Implemented** posture analysis for 2 ergonomic indicators (slouching and viewing distance) to provide real-time feedback during desk work
- **Built** distraction monitoring for 2 focus-related behaviors, detecting phone use and lack of interaction and triggering customizable alerts

Skills

CAD & Mechanical Design: SolidWorks, Onshape, ANSYS FEA, CFD, 3D Printing, CAD drawings
Manufacturing & Prototyping: CNC, Manual Milling, Mechanical Assembly, Prototyping, Design Verification, DFM, GD&T
Programming & Data Analysis: Data Structures and Algorithms, Python, C, MATLAB, NumPy, OpenCV, YOLOv11-OBB
Languages: Thai (Native), English (Fluent), Mandarin (Intermediate, HSK 4)
Relevant Coursework: Fluid Mechanics, Thermodynamics, Heat Transfer, Robot Kinematics & Dynamics, Principles of Imperative Computation